

Open Financial LLM Leaderboard

Jaisal Patel¹, Andy Zhu¹, Xiao-Yang Liu^{1,2}, Isabella Siu¹, Minna Huang¹, Akshaya Ramachandran¹, Kaiwen He¹, David Chiang¹

¹Rensselaer Polytechnic Institute, ²Columbia University



Overview

Background
The OpenFinLLM Leaderboard is a specialized platform designed to evaluate financial language models (LLMs). With the increasing complexity of financial language, there is a critical need for models tailored to the finance industry. This platform helps researchers and professionals identify the most effective LLMs for specific financial tasks by providing a transparent evaluation framework.

This semester, we accomplished:

- **Benchmarking:** Continued evaluating and benchmarking various models and new model versions on the OpenFinLLM Leaderboard on Hugging Face, assessing 10 models across 40 tasks each, and compiling the results into a detailed spreadsheet for comparison and analysis.
- **Tutorial:** Developed tutorials on using Google Colab scripts and explained the Zero-Shot functionality to streamline model testing and integration.
- **Documentation:** Worked on refining the user interface (UI) of the platform.
- **Future Goals:** Working towards creating a child/parent leaderboard system to connect multiple leaderboards across languages, aiming to calculate a greater average performance and enhance model evaluation by comparing cross-linguistic performance to improve overall accuracy.

Benchmarking Results

Model	Taiwan	Portoseguro	German	ACL18	BigData22	FiQA-SA
DeepSeek-R1-Distill-Llama-8B	0.9678	0.5344	0.6600	0.5027	0.5156	0.6681
DeepSeek-R1-Distill-Qwen-1.5B	0.9678	0.0647	0.6600	0.5250	0.5489	0.3574
Go4miii/DISC-FinLLM	0.9678	0.4912	0.6500	0.4933	0.4728	0.4766
NousResearch/Hermes-3-Llama-3.1-8B	0.9678	0.4362	0.3300	0.5062	0.5217	0.6851
Duxiaoman-DI/XuanYuan-6B	0.5993	0.5227	0.5000	0.5126	0.5476	0.5064
baichuan-inc/Baichuan-7B	0.9678	0.0651	0.6600	0.5013	0.5312	0.3617

Table 1: Sample Benchmarking Results for Several LLMs)

See <https://huggingface.co/spaces/finosfoundation/Open-Financial-LLM-Leaderboard> for more results.

T	Model	Average
1	GPT4-turbo	39.2
2	meta-llama/Llama-3.1-70B	36.2
3	Qwen/Qwen-72B	34.7
4	Duxiaoman-DI/XuanYuan-70B	34.4
5	meta-llama/Llama-3.1-8B	34.3
6	NousResearch/Hermes-3-Llama-3.1-8B	33.7
7	Gemini-1.0	32.4
8	utter-project/EuroLLM-9B-Instruct	31.7
9	GPT3.5-turbo	29.2
10	Qwen/Qwen2.5-7B	28.2
11	DeepseekV3	28
12	mistralai/Mistral-7B-Instruct-v0.3	26.6

Results meaning: We benchmark various LLMs on a diverse set of financial datasets—labeled here as “Taiwan,” “Portorsego,” “German,” “ACL18,” “BigData22,” and “FiQA-SA”, etc(40 more)—to ensure a broad evaluation of each model’s capabilities. By measuring performance across these datasets, we capture how well a given model handles different domains, languages, and topic areas. Once the individual metrics are collected, we compute an overall average score for each model. This average then becomes the model’s primary ranking criterion in the subsequent child leaderboard.



Child/Parent Leaderboard

Future Goal:

- Create a Parent Leaderboard to aggregate results from individual Child Leaderboards across languages (e.g., Greek, Chinese, etc.).
- Instead of averaging evaluation metrics, the Parent Leaderboard will focus on averaging the % performance scores of each language’s leaderboard.
- This will offer a more nuanced view of model performance by considering language-specific reliability.

Openness

Unique OpenFin LLM

- Focuses on community ideas and Openness.
- Unlike many closed leaderboards, everything is public and accessible.

Problems with Private Testing

- Hidden testing makes it hard to know if scores are fair.
- Limited tests can lead to scores that don’t really match performance.

Benefits of Open Leaderboard

- All test data is available for anyone to see, which builds trust.
- Experts from the community can suggest better ways to test.
- Open testing helps create scores that truly reflect model ability.

Fair Ranking Process

- Sharing details makes it easy for anyone to check the results.
- Regular feedback keeps the leaderboard updated.
- Involving many people helps reduce bias in the tests.

Future Testing Methods

- Testing can quickly adapt to new trends and ideas.
- Working together leads to better, long-lasting testing.
- Open access lets everyone help improve the system.

Documentation

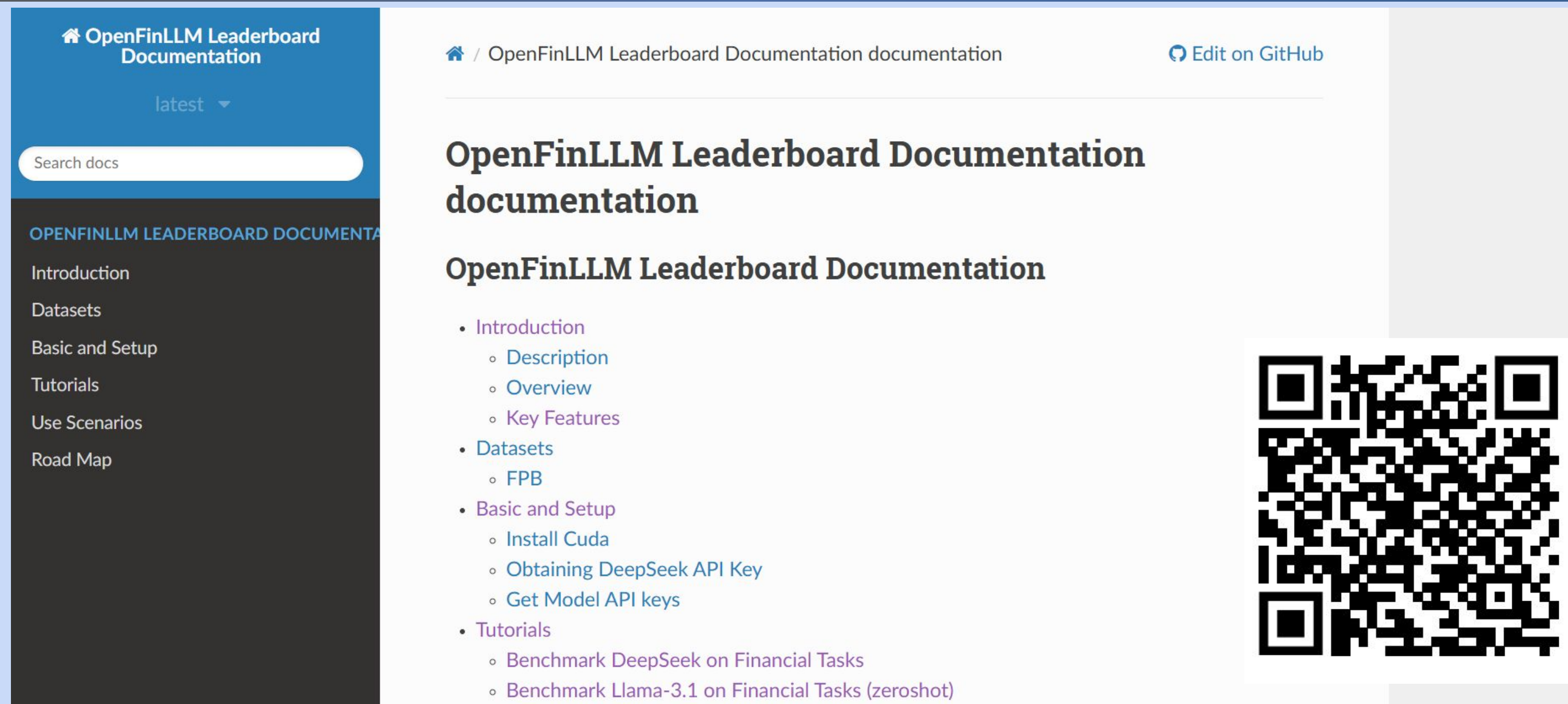


Figure 2: OpenFin LLM Leaderboard Documentation Website

Introduction: Provides a general overview of OpenFin LLM Leaderboard, including purpose and key features

Datasets: Details the datasets used within the project, such as FPB datasets

Basic and Setup: Guides users through the initial setup process, including installing CUDA and obtaining necessary API keys

Tutorials: Offers step-by-step tutorials for benchmarking variou smodels (e.g., DeepSeek, Llama-3.1, ChatGPT) on financial tasks

Use Scenarios: Explores practical applications of the project, such as financial document analysis and AI-driven regulatory compliance

Road Map: Outlines the future plan and development roadmap for the OpenFin LLM Leaderboard.

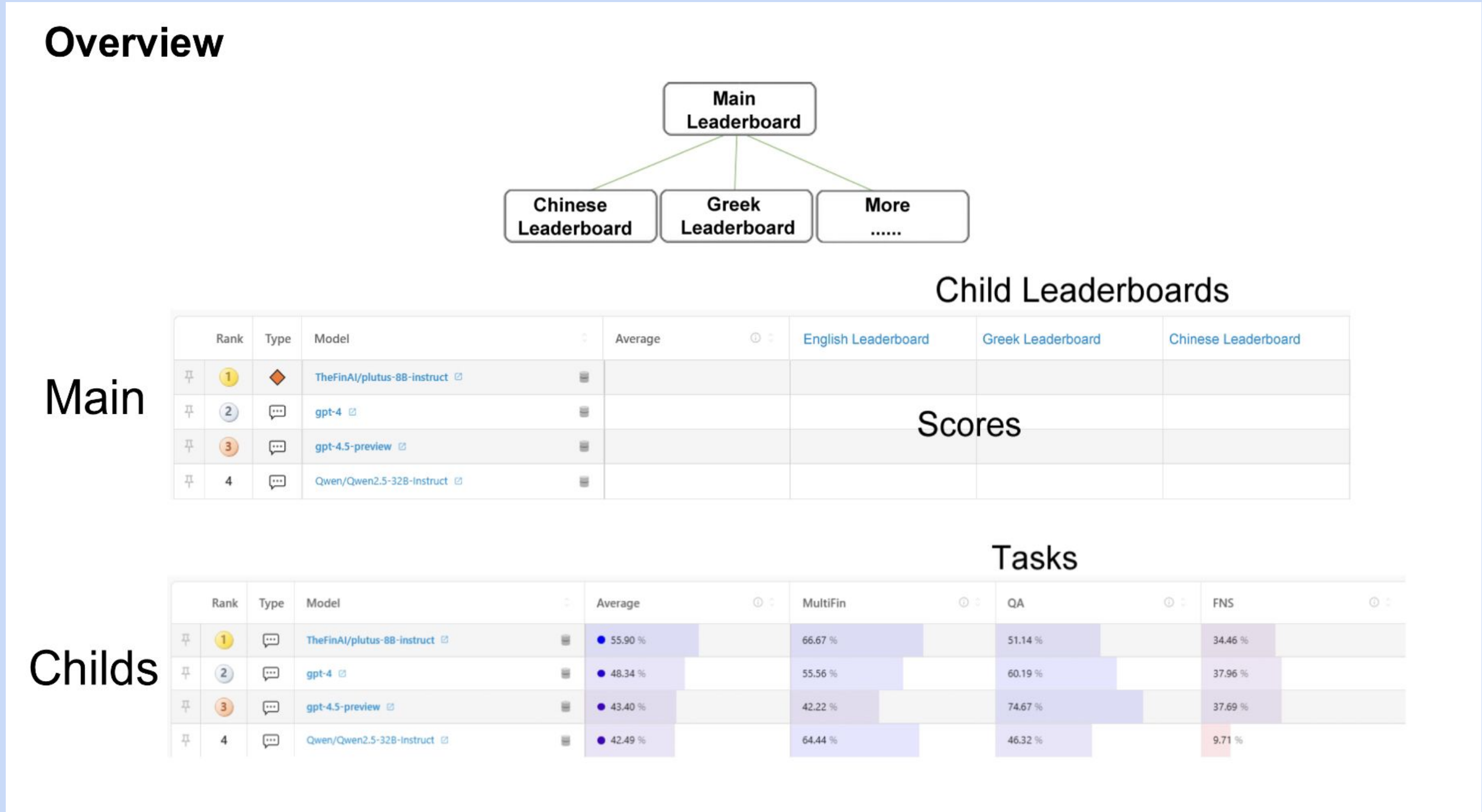


Figure 1: OpenFin LLM Leaderboard Child-Parent Structure